Anonymity Testing With Tor And Proxychains

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Executive Summary

This lab focused on testing anonymity of users while navigating the internet using tor and proxychains. Key activities include observation of tor and proxychains operations and persistence with regards to ip address routing under specified conditions. The lab revealed that the proxy chain relies on trusting the previous and next hops while tor has no idea of the previous or next hop which completely masks the first hop or user ip address.

1.0 Introduction

Anonymity testing with Tor and Proxychains refers to the systematic evaluation of techniques used to conceal the origin of network traffic by routing communications through intermediary nodes. Tor provides a layered encryption network designed to anonymize traffic, while Proxychains enforces proxy-based routing at the application level. Studying their combined use helps examine anonymity effectiveness, traffic obfuscation, and potential information leakage, making it relevant for privacy, network security, and adversarial traffic analysis.

2.0 Aim and objectives

The aim of this lab is to explore, compare and validate the operations of Tor and proxy chains in routing and anonymity consistency.

The Objectives of this lab includes;

- Installation and configuration of Tor and proxy chains to route traffic through the Tor network
- Verify anonymity by comparing real ip addresses with Tor exit nodes ip address
- Analyze Tor circuit persistency and IP rotation behavior
- Identify limitations and risk of using Tor and proxychains during security testing

2.2 Tools

- Tor
- Proxychains
- Curl
- Firefox

3.0 Methodology

3.1 Tor

- The Linux terminal was opened and Tor was started with the command ***Sudo service tor start*** followed by a confirmation with ***sudo Service tor status***
- The proxychain was configured for anonymity using the nano text editor at ***/etc/proxychains.conf.***
- Key configuration includes the ***strict_chain, proxy_dns, tcp_read_time_out 15000, tcp_connect_time_out 8000; [proxyList] sock5 127.0.0.1 9050.***
- Strict chain ensured traffic passed through Tor while proxy dns prevented traffic leak.
- The command ***curl*** https://httpbin.org/ip was run with a response of "origin": "**102.89.33.55**"
- The response IP matches the local network ip address from an ISP.

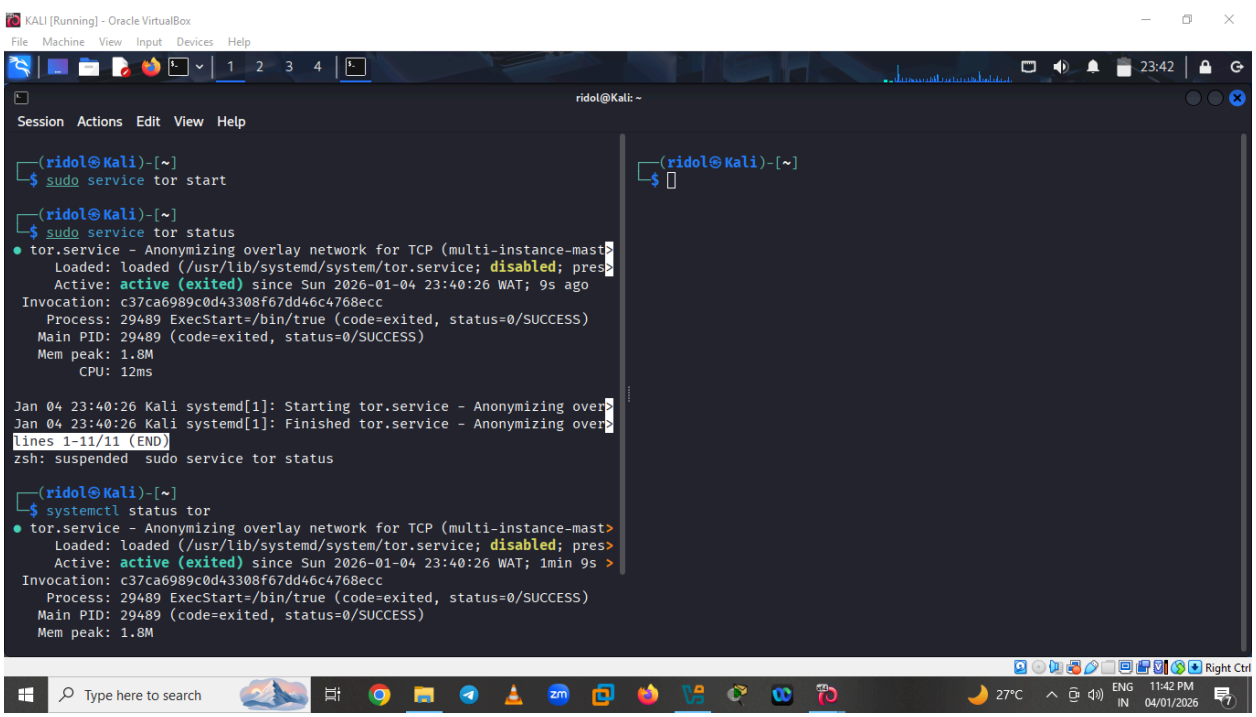

Fig 1: Evidence showing the Tor service activated from the Linux terminal

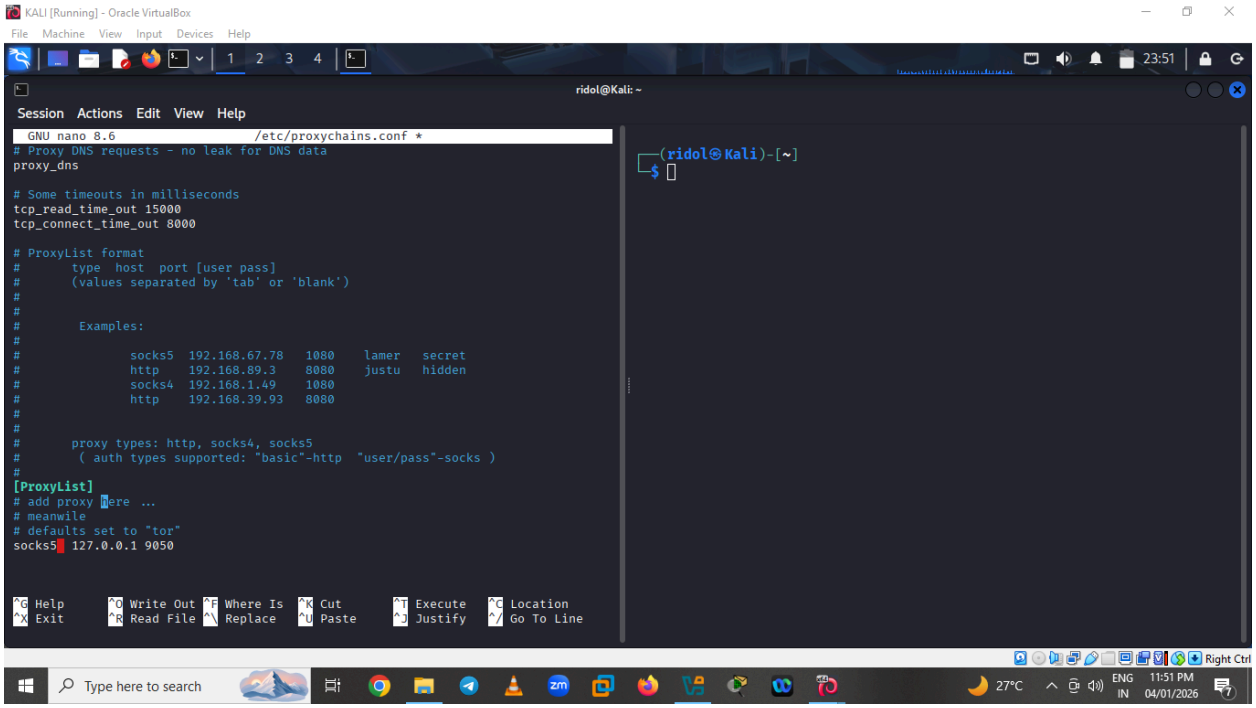

Fig 2: Evidence showing the configuration of proxychain on nano text editor.

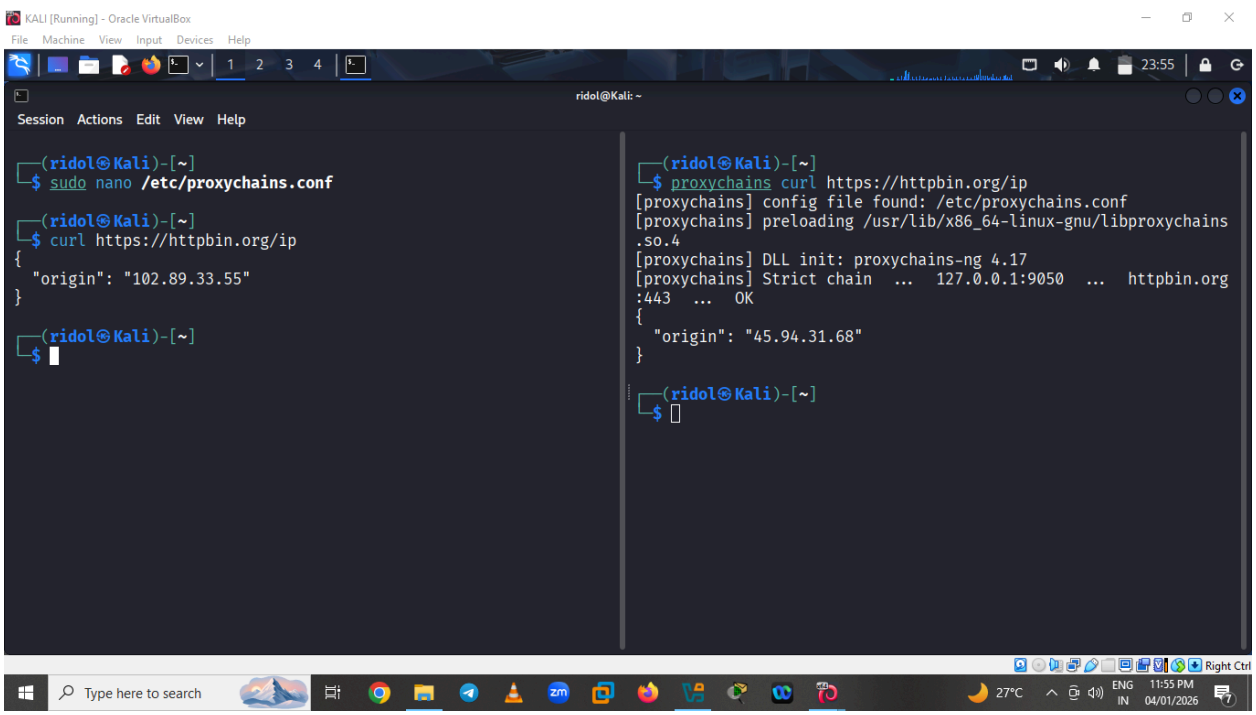

Fig 3: Evidence showing the Ip address from the local network ISP

3.2 Proxychains

- The proxy chains was launched using the command ***proxychains curl*** ***https://httpbin.org/ip*** with a response of "***origin": "45.94.31.68***"
- The command was run several time and the same ip address was returned indicating a persistent behavior with tor circuit reuse when combined with proxychain
- To initiate a new identity for tor the command sudo service tor restart was run and the proxy chain command ***proxychains curl*** ***https://httpbin.org/ip*** was repeated
- A new ip address ***185.220.101.43*** was returned and persistent upon several attempts.
- A firefox browser was opened for web based ip confirmation through proxy chain using the command proxychains firefox
- The same ip address ***185.220.101.43*** was confirmed from the website ***whatismyip.com***

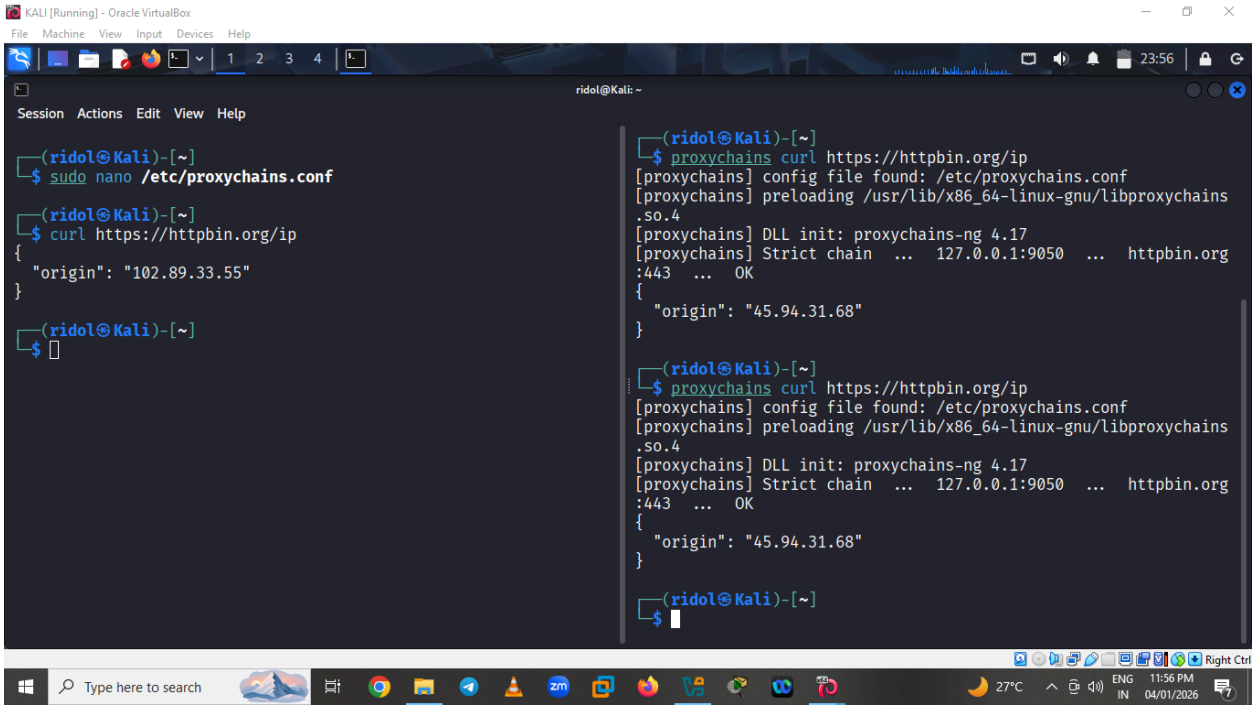

Fig 4: Evidence showing a persistent ip address from proxychain

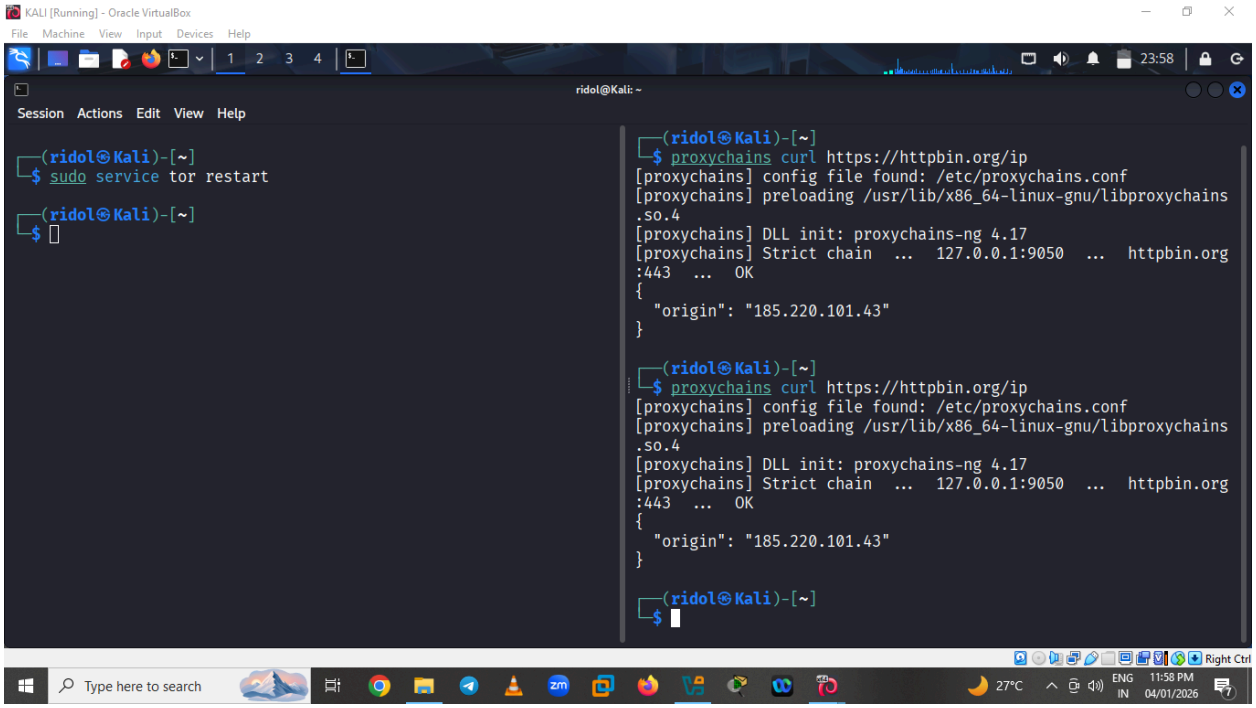

Fig 5: Evidence showing a different ip address from the proxychain upon tor restart

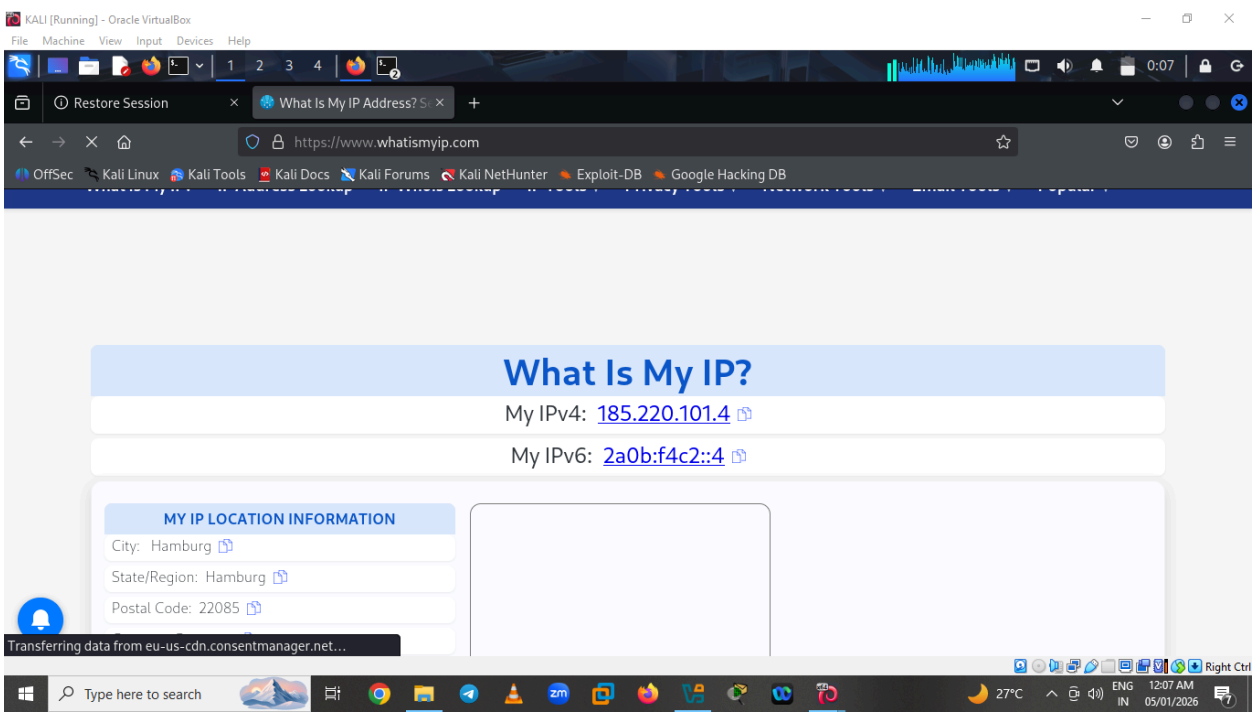

Fig 6: Evidence showing confirmation of IP address from web application whatismyip.com

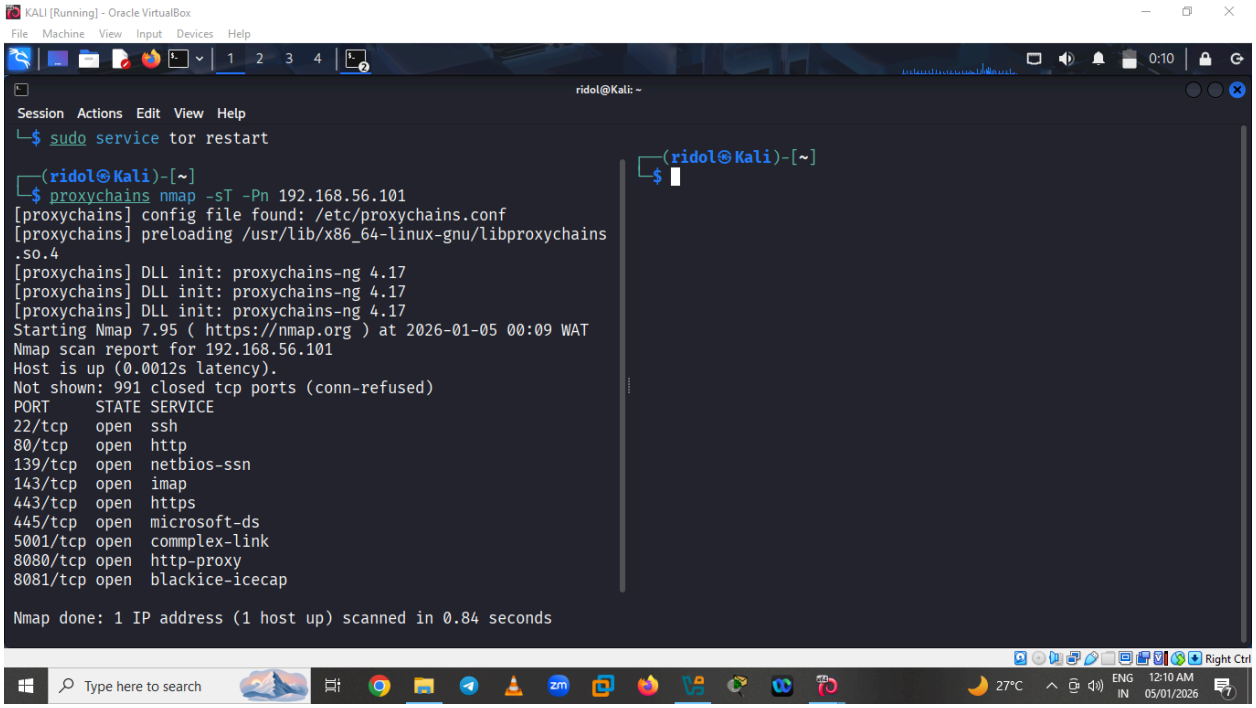

Fig 7: Evidence showing the results from network scan through proxy chain

4.0 Results and Findings

4.1 Strict chain, dynamic chain and random chain

A strict chain proxy type ensures that traffic is forced through all proxies in a fixed order, resulting in a consistent exit IP but no tolerance for proxy failure.

Dynamic chain proxy tends to be slightly different from the strict chain in such that traffic follows the proxy list in order but skips unavailable proxies, providing moderate IP consistency with improved reliability.

The random chain proxy is completely different from the rest. Traffic is routed through randomly selected proxies for each connection, causing frequent exit IP changes and lower predictability

4.2 Proxychains Anonymity and Tor persistent circuit behavior

Proxychains provides anonymity by redirecting traffic through one or more proxy servers, masking the original source IP with the exit proxy's IP.

However, anonymity is limited and conditional, because the exit IP remains consistent as long as the same proxy chain is used, making traffic correlation and logging by proxy operators possible. This confirms a clear redirection of traffic from the isp ip address 102.89.33.55 to proxy ip 45.94.31.68 from the lab.

Tor uses temporary but persistent circuits, where traffic is routed through the same set of relays for a defined period before rotation.
During this time, the exit IP remains stable, supporting session continuity, but is periodically changed to reduce long-term tracking and correlation.

4.3 Tor persistency and Anonymity Importance

Tor does not change exit IPs for every request because it uses persistent circuits for a short time. Multiple requests may exit through the same Tor exit IP until the circuit is rotated.

Consistency in anonymity testing is important because it enables reliable, repeatable observations that help distinguish intentional anonymity behavior from misconfigurations, leaks, or random network variation.

5.0 Conclusion

The onion router and proxychain plays a vital role in anonymity and privacy of users while maintaining ethical behavior on the internet. Different modes of proxychain exhibit different operational behavior.
It's important to note that Tor protects network identity, not user identity, and anonymity is lost when traffic leaks or personal authentication is introduced.